

brands? The clock is ticking, say experts. Is India prepared to compete and fill the vacuum created by the world's technological overdependence on our neighbour?

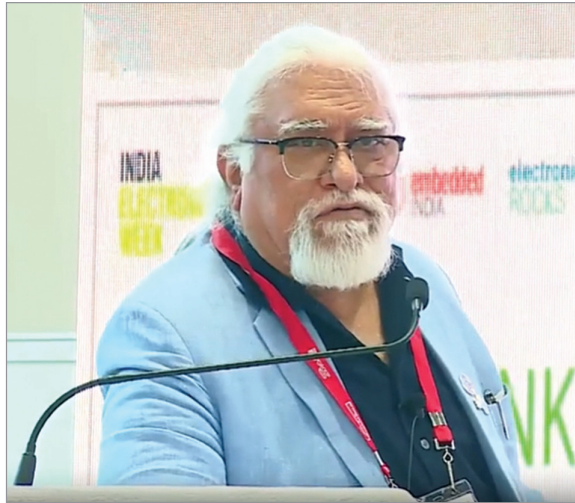
We have barely scratched the surface

Though relatively young, India's ESDM industry has a long history. Initially dominated by public sector undertakings. The sector opened up post-liberalisation, with private Indian firms and global players entering the market. The first significant investments occurred between 2005 and 2007, when companies such as Nokia and Jabil Circuits established operations.

"I think the first time the term ESDM was coined was back in the 2008-09 time frame when, along with several other industry leaders, we were working on a recommendation for the government of India that subsequently, in 2011-12 time frame, resulted in the first national policy in electronics," recollected Jaswinder Ahuja, ex-Corporate Vice President of Cadence Design Systems.

Despite a temporary slowdown in 2013-14, the sector rebounded, and today, India is a key global destination for electronics manufacturing. The sector operates through four primary business models. The most common is EMS (electronic manufacturing services), where original equipment manufacturers provide the design and ESDM firms handle the manufacturing. The ODM (original design manufacturing) model goes further, with ESDM firms managing everything from product design to development and production.

On the other hand, job work involves short-term, contract-based manufacturing or prototyping, often serving as an entry point for longer engagements. Lastly, after-sales services, including repair, refurbishment, and maintenance, are gaining



N. Kishor Narang, Founder, Narnix Technologies

importance as companies aim to enhance brand loyalty and address growing concerns around e-waste and product lifecycle support.

The industry currently comprises over 700 companies, including global giants (Bharat FIH, Wistron, and Pegatron), large Indian players (Dixon, Kaynes, and Syrma), and mid-sized firms (VVDN and Avalon). The market is dominated by mobile phones, consumer electronics, and industrial electronics, which collectively contribute over 75% of the total output. The fact is, EMS still accounts for 80% of the market, and ODM is still a growing segment—rather one that is yet to grow. The initial government push was towards manufacturing.

However, manufacturing without design would not be a good road map for long-term success. N. Kishor Narang, the founder of Narnix Technolabs, reminded us that we first made India the back office of the world by opening a multitude of BPOs, and by simply setting up an EMS plant without indigenous R&D for Chinese and global brands, we are turning India into the backyard factory of the world.

Citing global trends, Jaswinder added, "Nvidia has no manufacturing

at all. It is all design; it is all intellectual property." In the case of India, despite having "the largest workforce of designers," the country lacks a true indigenous design ecosystem—most talent operates in global capability centres. It is a "huge vacuum" that must be addressed. Although there are schemes like ISM (India Semiconductor Mission), C2S (Chips to Startup), and DLI

(Design-Linked Incentive Scheme), the offtake has been slow, which all the more requires private enterprise to step up. "The government can only do so much... private enterprise has to step forward," stated Jaswinder.

A peep into India's policy playbook, powering ESDM growth

Looking at how much the government has done, key efforts include the Production Linked Incentive (PLI) schemes, with ₹15.96 billion disbursed in H1 FY25 and targeted extensions to FY26, especially for electronics, IT hardware, and large-scale manufacturing. The highly commended ISM, with an initial ₹760 billion outlay, and the recent foundation of three semiconductor plants (₹1.25 trillion investment) highlight India's focus on chip manufacturing.

According to the *India Brand Equity Foundation (IBEF)*, partnerships, such as India-Taiwan talks and a proposed \$7.5 billion fab facility, aim to boost chip production. Budget allocations for MeitY (Ministry of Electronics and Information Technology) have seen major increases too: ₹143 billion in FY23 and ₹165.49 billion in FY24.

The National Policy on Electronics